

Past Problems

for Korean Undergraduate Mathematics Competition

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This document collects past problems from the Korean Undergraduate Mathematics Competition. General mathematical notation and writing follow *(MATH000) Mathematical Language*.

2001.2.1. Let $0 < a, b < 1$ satisfy

$$\int_0^a \frac{1}{\sqrt{1-x^2}} dx = \int_b^1 \frac{1}{\sqrt{1-x^2}} dx.$$

Find $a^2 + b^2$.

2001.2.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

2001.2.3. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2y = 0, \quad x > 0.$$

Hint: $y = e^{x^2}$ is a solution of the differential equation.

2001.2.4. Let $f: [0, \infty) \rightarrow [0, \infty)$ satisfy

$$2f(x)f(1) \leq f(x)f(1) + 1 \leq 2f(2x)$$

for every $x \geq 0$. Prove that f is constant.

2001.2.5. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative. Evaluate

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(x) \left(|\sin(\lambda x)| - \frac{2}{\pi} \right) dx.$$

2001.2.6. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C , respectively. Find the maximum value of

$$a \cos A + b \cos B + c \cos C.$$

2001.2.7. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.2.8. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ be a unit vector. Prove that

$$\sum_{i=1}^n \|\mathbf{x} - \mathbf{v}_i\|^2 \geq 2n - 2\sqrt{n}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

2001.2.9. Let $\lambda \geq 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2001.1.1. Show that there are infinitely many triples $(a, b, c) \in \mathbb{N}^3$ satisfying

$$a \cos\left(\arctan \frac{b}{a}\right) + b \sin\left(\arctan \frac{b}{a}\right) = c.$$

2001.1.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

2001.1.3. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2y = 0, \quad x > 0.$$

Hint: $y = e^{x^2}$ is a solution of the differential equation.

2001.1.4. Let $f: [0, \infty) \rightarrow [0, \infty)$ satisfy

$$2f(x)f(1) \leq f(x)f(1) + 1 \leq 2f(2x)$$

for every $x \geq 0$. Prove that f is constant.

2001.1.5. Evaluate

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} \frac{\sin x}{n^2 \left(x - \frac{\pi}{4}\right)^2 + 1} dx.$$

2001.1.6. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C , respectively. Find the maximum value of

$$a \cos A + b \cos B + c \cos C.$$

2001.1.7. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^2$ satisfy

$$\|\mathbf{v}_i\| \leq 1 \quad (i = 1, 2, \dots, n).$$

Prove that there always exist signs

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{1, -1\}$$

such that

$$\|\varepsilon_1 \mathbf{v}_1 + \varepsilon_2 \mathbf{v}_2 + \dots + \varepsilon_n \mathbf{v}_n\| \leq \sqrt{2}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

2001.1.9. Let $\lambda \geq 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2002.0.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Suppose the function $x \mapsto xf(x)$ is differentiable for $x \neq 0$ and

$$\frac{d}{dx}(xf(x)) = e^x.$$

Find f .

2002.0.2. Let (a_n) be an arithmetic sequence with first term a and common difference d , and define the 2002×2002 matrix

$$\mathbf{A} := \begin{pmatrix} a_1 & a_2 & a_3 & a_4 & \cdots & a_{2002} \\ a_2 & a_1 & a_2 & a_3 & \cdots & a_{2001} \\ a_3 & a_2 & a_1 & a_2 & \cdots & a_{2000} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2002} & a_{2001} & a_{2000} & a_{1999} & \cdots & a_1 \end{pmatrix}.$$

Express $\det(\mathbf{A})$ in terms of a and d .

2002.0.3. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be unit vectors in $\mathbb{R}^n$. Prove that

$$|\langle \mathbf{a}, \mathbf{c} \rangle \langle \mathbf{b}, \mathbf{d} \rangle - \langle \mathbf{a}, \mathbf{d} \rangle \langle \mathbf{b}, \mathbf{c} \rangle| \leq 1,$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^n$.

2002.0.4. Let $n \in \mathbb{N}$, and let $a_1, \dots, a_n, x_1, \dots, x_n$ be positive real numbers satisfying

- (i) $a_1 \leq a_2 \leq \dots \leq a_n$;
- (ii) $x_k \geq x_n$ for $k = 1, 2, \dots, n-1$.

Prove that

$$(a_1 + x_1 a_2)(a_2 + x_2 a_3) \cdots (a_{n-1} + x_{n-1} a_n)(a_n + x_n a_1) \geq a_1 a_2 \cdots a_n (x_1 + 1)(x_2 + 1) \cdots (x_n + 1).$$

2002.0.5. For $\alpha \in \mathbb{C}$, define the $n \times n$ matrix $\mathbf{A}_n = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & i = j, \\ \alpha, & i - j = -1, \\ \bar{\alpha}, & i - j = 1, \\ 0, & |i - j| \geq 2. \end{cases}$$

Prove that $\mathbf{A}_n$ is positive semidefinite for every $n = 1, 2, \dots$ if and only if

$$|\alpha| \leq \frac{1}{2}.$$

Here $\bar{\alpha}$ denotes the complex conjugate of α .

2002.0.6. Solve the system of differential equations

$$x''(t) + 2y'(t) - x(t) = 0, \quad y''(t) - 2x'(t) - y(t) = 0$$

subject to

$$x(0) = y(0) = 0, \quad x'(0) = a > 0, \quad y'(0) = 0.$$

2002.2.1. For $(x, y) \neq (0, 0)$, define

$$f(x, y) = \frac{x^2 - xy + y^2}{2x^2 + y^2}.$$

Find the minimum value of f .

2002.2.2. Let $f, g, h: [-1, 1] \rightarrow \mathbb{R}$ be continuous. Suppose that f is odd, g is even, and

$$\int_{-1}^1 f(x)^2 \, dx = \int_{-1}^1 g(x)^2 \, dx = 1.$$

Prove that

$$\left(\int_{-1}^1 f(x)h(x) \, dx \right)^2 + \left(\int_{-1}^1 g(x)h(x) \, dx \right)^2 \leq \int_{-1}^1 h(x)^2 \, dx.$$

2002.2.3. Let $a > b > 0$. Evaluate

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx.$$

2002.1.1. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative, with $f(a) = 0$. Suppose that

$$\left| \frac{f'(x)}{2003} \right| \leq \left| \frac{f(x)}{2002} \right| \quad (x \in [a, b]).$$

Prove that $f(x) \equiv 0$ on $[a, b]$.

2002.1.2. Let $n \in \mathbb{N}$ with $n > 1$. Prove that

$$\frac{((n-1)!)^{1/(n-1)}}{(n!)^{1/n}} < \frac{(n!)^{1/n}}{((n+1)!)^{1/(n+1)}}.$$

It is given that

$$4 > e^{4-2\sqrt{2}}.$$

2002.1.3. Let

$$\alpha = 0.0110101000101000101 \dots,$$

where the n th digit after the decimal point is 1 if n is prime and 0 otherwise. Prove that α is irrational.

2003.2.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2003.2.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC , BCD , CDA , and DAB , respectively. Prove that $PQRS$ is a rectangle.

2003.2.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.2.4. Let (x_n) be a sequence satisfying

$$0 \leq x_{n+1} \leq x_n$$

for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} x_n$$

converges. Determine

$$\lim_{n \rightarrow \infty} nx_n.$$

2003.2.5. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ satisfy

$$x_1 + x_2 + \dots + x_n = 0.$$

Prove that

$$\max_{1 \leq j \leq n} |x_j| \leq \left(\frac{n-1}{n} \right)^{1/2} \|\mathbf{x}\|_2,$$

where

$$\|\mathbf{x}\|_2 = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}.$$

2003.2.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.2.7. Prove that the real number $\sin 1$ is irrational.

2003.2.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3} \right)^n (n+1)$$

for every $x \in A$.

2003.2.9. Let f be the 2π -periodic function defined by

$$f(x) = \begin{cases} \pi + x, & -\pi < x \leq 0, \\ \pi - x, & 0 < x \leq \pi. \end{cases}$$

Using the Fourier series of f , prove that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \dots + \frac{1}{(2n-1)^2} + \dots.$$

2003.1.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2003.1.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC, BCD, CDA , and DAB , respectively. Prove that $PQRS$ is a rectangle.

2003.1.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.1.4. Let (a_n) be a sequence with $a_n \geq 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n$$

converges. Prove that, for every $p > \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

converges. Also give an example showing that, when $p = \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

may diverge.

2003.1.5. Let $p \geq 1$ be a real number, and let $x_1, x_2, \dots, x_n > 0$ satisfy

$$x_1 + x_2 + \dots + x_n = 1.$$

Prove that

$$\sum_{j=1}^n \left(x_j + \frac{1}{x_j} \right)^p \geq \frac{(1 + n^2)^p}{n^{p-1}}.$$

2003.1.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

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2003.1.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3}\right)^n (n+1)$$

for every $x \in A$.

2003.1.9. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix whose entries are all 0 or 1. Suppose that, for some $t \in \mathbb{N}$,

$$\mathbf{A}\mathbf{A}^T = \begin{pmatrix} t+1 & 1 & \cdots & 1 & 1 \\ 1 & t+1 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & t+1 & 1 \\ 1 & 1 & \cdots & 1 & t+1 \end{pmatrix} = t\mathbf{I}_n + \mathbf{J},$$

where $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$, $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\mathbf{J}$ is the $n \times n$ all-ones matrix. Express $\det(\mathbf{A})$ as a function of t .

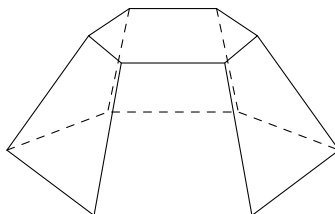
2004.2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. Prove the following.

- (a) If f is continuous at 0, then f is continuous at every $x \in \mathbb{R}$.
- (b) If f is continuous on $\mathbb{R}$ and $f(1) = c$, then $f(x) = cx$ for every $x \in \mathbb{R}$.

2004.2.2. A frustum of a pyramid has parallel upper and lower bases and uniform density. Find the ratio of the distance from its center of mass to the lower base to the distance from its center of mass to the upper base. The similarity ratio of the upper base to the lower base is r , where $r < 1$.



2004.2.3. A convex quadrilateral $ABCD$ has side lengths a, b, c, d , in this order starting from AB . Show that if its area is maximal among all such quadrilaterals, then $ABCD$ is cyclic.

2004.2.4. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$. Find the orthogonal complement of

$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.2.5. Let $f: [a, c] \rightarrow (0, \infty)$ be continuous on $[a, c]$ and differentiable on (a, c) . Let

$$P = (x_0, f(x_0))$$

be a point on the graph of f . Let α and γ be the angles made with the x -axis by the lines joining P to $(a, 0)$ and $(c, 0)$, respectively, and let β be the angle made with the x -axis by the tangent line to the graph at P . Show that there exists $x_0 \in (a, c)$ such that

$$\tan \alpha + \tan \beta + \tan \gamma = 0.$$

If the tangent line does not meet the x -axis, take $\beta = 0$.

2004.2.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

$$\sum_{n=1}^{\infty} a_n^2$$

converges. If

$$\sum_{n=1}^{\infty} f(a_n)$$

converges, prove that

$$\sum_{n=1}^{\infty} |f(a_n)|$$

also converges.

2004.2.7. Suppose a plane curve $(x(t), y(t))$ of period T satisfies

$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy, \quad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.2.8. Evaluate

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx.$$

2004.2.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

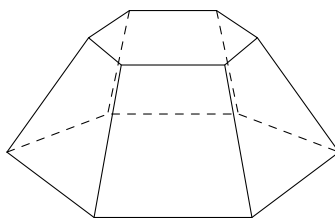
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$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.1.5. Let A be the least positive integer of the form

$$33^k - 7^\ell, \quad k, \ell \in \mathbb{N},$$

and let B be the least positive integer of the form

$$7^m - 33^n, \quad m, n \in \mathbb{N}.$$

Find $A + B$.

2004.1.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

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$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy, \quad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.1.8. Let $\mathbf{v} \in \mathbb{R}^n$ be a unit vector, and let $\mathbf{P}_{\mathbf{v}}$ denote the orthogonal projection onto the line $\text{span}(\mathbf{v})$, so that

$$\mathbf{P}_{\mathbf{v}}\mathbf{x} = (\mathbf{x} \cdot \mathbf{v})\mathbf{v}$$

for every $\mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{A}$ be an $n \times n$ positive definite symmetric matrix. Find the maximum value of λ such that

$$\mathbf{A} - \lambda \mathbf{P}_{\mathbf{v}} \succeq 0.$$

Here $\mathbf{B} \succ 0$ means that every eigenvalue of the symmetric matrix $\mathbf{B}$ is positive, and $\mathbf{B} \succeq 0$ means that every eigenvalue is nonnegative.

2004.1.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

2005.0.1. Let

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & \sqrt{2} \\ -\sqrt{2} & 0 \end{pmatrix}.$$

Find the matrix

$$e^{\mathbf{A}} = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!}.$$

2005.0.2. Solve the differential equation

$$y'(t) + 2ty(t) = ty(t)^2, \quad y(1) = 1.$$

2005.0.3. Let $p \geq 3$ be an arbitrary prime, and let

$$\mathbb{F}_p = \{0, 1, \dots, p-1\}.$$

Show that the set

$$S = \{(x, y) \in \mathbb{F}_p^2 \mid x^2 + y^2 \equiv -1 \pmod{p}\}$$

is nonempty.

2005.0.4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that

$$|f(n)| \leq \frac{1}{(1+n)^{5/4}}$$

for every nonnegative integer n , and that

$$|f'(x)| \leq e^{-x}$$

for every $x \in \mathbb{R}$. Prove that

$$\int_0^{\infty} |f(x)| \, dx < \infty.$$

2005.0.5. Let S_n be the set of all bijections from $\{1, 2, \dots, n\}$ to itself. For $\sigma \in S_n$, let $f(\sigma)$ denote the number of fixed points of σ , that is,

$$f(\sigma) = |\{x \in \{1, 2, \dots, n\} \mid \sigma(x) = x\}|.$$

Find

$$\sum_{\sigma \in S_n} f(\sigma).$$

2005.2.1. A population consists of smokers and nonsmokers. Each year, 5% of the nonsmokers begin smoking and 10% of the smokers quit smoking. Prove that, after a sufficiently long time, the ratio of smokers to nonsmokers converges to a constant, and find that ratio.

2005.2.2. Let X be the set of all invertible 2×2 matrices and Y the set of all invertible 3×3 matrices. Define $\Phi: X \rightarrow Y$ by

$$\Phi \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

Suppose $\mathbf{A} \in X$ has eigenvalues $1, x$ and $\mathbf{B} \in X$ has eigenvalues $1 + x, 1 - x$. Find

$$\det(\Phi(\mathbf{AB})).$$

2005.2.3. Words of length n are formed using the three letters A, B, C , with the restriction that A may not occur twice consecutively. Express the number of such words as a function of n .

2005.2.4. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be twice differentiable and suppose that

$$|f(x)| \leq 1, \quad |f''(x)| \leq 1$$

for every $x \in (0, \infty)$. Prove that

$$|f'(x)| \leq 2$$

for every $x \in (0, \infty)$.

2005.1.1. A population consists of smokers and nonsmokers. Each year, 5% of the nonsmokers begin smoking and 10% of the smokers quit smoking. Prove that, after a sufficiently long time, the ratio of smokers to nonsmokers converges to a constant, and find that ratio.

2005.1.2. Let $f: [0, 1] \rightarrow \mathbb{R}$ be differentiable, with $f(0) = 0$ and $f(1) = 1$. Prove that, for every $n \in \mathbb{N}$, there exist distinct points $x_1, x_2, \dots, x_n \in [0, 1]$ such that

$$\sum_{k=1}^n \frac{1}{f'(x_k)} = n.$$

2005.1.3. Words of length n are formed using the three letters A, B, C , with the restriction that A may not occur twice consecutively. Express the number of such words as a function of n .

2005.1.4. Let (a_n) and (b_n) be sequences satisfying $a_n > 0$ and $b_n > 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n = 3, \quad \sum_{n=1}^{\infty} b_n = 4.$$

Find the supremum and infimum of all possible values of

$$\sum_{n=1}^{\infty} \frac{a_n^2}{a_n + b_n}.$$

2006.0.1. Let $f(x) = \arctan(2x)$. Find $f^{(99)}(0)$.

2006.0.2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous periodic function with period $2T$. Prove that there exists a suitable $x_0 \in \mathbb{R}$ such that

$$f(x_0 + T) = f(x_0).$$

2006.0.3. Let $\mathbf{A}$ be an $n \times n$ matrix such that

$$\text{tr}(\mathbf{X}\mathbf{Y}\mathbf{A}) = \text{tr}(\mathbf{Y}\mathbf{X}\mathbf{A})$$

for every pair of $n \times n$ matrices $\mathbf{X}, \mathbf{Y}$. Prove that $\mathbf{A}$ is a scalar multiple of the identity matrix. Here tr denotes the sum of the diagonal entries.

2006.0.4. Let (a_n) be a sequence of positive real numbers, and suppose that the limit

$$\lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right)$$

exists. Prove that if this limit is greater than 1, then the series

$$\sum_{n=1}^{\infty} a_n$$

converges.

2006.0.5. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be differentiable, suppose that f' is continuous, and assume that

$$\lim_{x \rightarrow \infty} (f(x) + f'(x)) = 0.$$

Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

2006.2.1. Evaluate

$$\int_{-\pi}^{\pi} \ln \left| \sin \frac{\theta}{2} \right| d\theta.$$

2006.2.2. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$ be a unit vector, and let $\mathbf{X} = (x_{ij})$ be the $n \times n$ matrix defined by

$$x_{ij} = \delta_{ij} + a_i a_j,$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Compute $\det(\mathbf{X})$.

2006.2.3. Let $h: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\int_0^{\infty} h(t) dt = 1.$$

Evaluate

$$\iint_{\mathbb{R}^2} h(x^2 + 2xy + 5y^2) dA.$$

2006.2.4. Evaluate the series

$$1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \cdots + \frac{1}{6n-5} - \frac{1}{6n-1} + \cdots.$$

2006.1.1. Prove that the improper integral

$$\int_0^\infty \sin(x^2) \, dx$$

converges, whereas

$$\int_0^\infty |\sin(x^2)| \, dx$$

diverges.

2006.1.2. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$ be a unit vector, and let $\mathbf{X} = (x_{ij})$ be the $n \times n$ matrix defined by

$$x_{ij} = \delta_{ij} + a_i a_j,$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Compute $\det(\mathbf{X})$.

2006.1.3. Let $h: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\int_0^\infty h(t) \, dt = 1.$$

Evaluate

$$\iint_{\mathbb{R}^2} h(x^2 + 2xy + 5y^2) \, dA.$$

2006.1.4. Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots.$$

Let $\sum a_k$ be a series obtained by rearranging its terms subject to the rule that the relative order among terms of the same sign is not changed. For example,

$$1 + \frac{1}{3} - \frac{1}{2} - \frac{1}{4} + \cdots$$

is allowed, whereas

$$1 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} + \cdots$$

is not. Among $a_1, a_2, \dots, a_n$, let p_n be the number of positive terms and q_n the number of negative terms.

(a) Prove that $\sum a_k$ converges if and only if the limit

$$\lim_{n \rightarrow \infty} \frac{p_n}{q_n}$$

exists.

(b) If this limit is α , prove that

$$\sum a_k = \ln 2 + \frac{1}{2} \ln \alpha.$$

2007.2.1. Evaluate

$$\int_0^1 \int_0^x (3 - x - y) \, dy \, dx.$$

2007.2.2. Let $C(t) = (\cos t, \sin t)$ for $0 \leq t \leq 2\pi$. Evaluate the line integral

$$\int_C \frac{y \, dx - x \, dy}{x^2 + y^2}.$$

2007.2.3. Find the area of the quadrilateral whose vertices are

$$(0, 0, 0), \quad (1, 2, 3), \quad (3, 1, 2), \quad (7, 4, 7).$$

2007.2.4. Let

$$f(x) = \frac{e^x}{x}, \quad 1 \leq x \leq 2,$$

and let g be the inverse function of f . Evaluate

$$\int_e^{e^2/2} [g(x)]^2 \, dx.$$

2007.2.5. Find the functions $x(t)$ and $y(t)$ satisfying

$$x'(t) = -y(t), \quad y'(t) = x(t), \quad (x(0), y(0)) = (1, 1).$$

2007.2.6. Find the relation among a, b, c that is necessary and sufficient for the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= a, \\ 2x_1 + 4x_2 + 6x_3 - 2x_4 &= b, \\ x_1 + x_3 + x_4 &= c \end{aligned}$$

to have a solution.

2007.2.7. Write the matrix

$$\begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{pmatrix}$$

as $\mathbf{LU}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

2007.2.8. Three unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ determine a parallelepiped of volume $\frac{1}{2}$. Let θ_{ij} be the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$, and let $\mathbf{A}$ be the 3×3 matrix whose (i, j) -entry is $\cos \theta_{ij}$. Compute $\det(\mathbf{A})$.

2007.2.9. Find the volume of the region in $\mathbb{R}^3$ described by

$$x = (p - q) \cos t, \quad y = (p - q) \sin t, \quad z = p + q,$$

where

$$0 \leq t \leq 2\pi, \quad 1 - \varepsilon \leq p \leq 1 + \varepsilon, \quad -\delta \leq q \leq \delta, \quad \delta < 1 - \varepsilon.$$

2007.2.10. Find the constants a, b that minimize

$$\int_0^\pi [\sin x - (ax + b)]^2 dx.$$

2007.2.11. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ be linearly independent vectors satisfying

$$\|\mathbf{v}_i\| = 1, \quad \mathbf{v}_i \cdot \mathbf{v}_j = -\frac{1}{3} \quad (i \neq j).$$

For vectors $\mathbf{x} \in \mathbb{R}^3$ with $\|\mathbf{x}\| = 1$, find the maximum and minimum values of

$$\sum_{i=1}^3 \|\mathbf{x} - \mathbf{v}_i\|^2.$$

2007.2.12. For real y , evaluate

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i y x}}{\cosh(\pi x)} dx.$$

2007.2.13. Let $p > 0$ be real. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n (1^p + 2^p + 3^p + \cdots + k^p)^{1/(p+1)}.$$

2007.1.1. Find the area of the quadrilateral whose vertices are

$$(0, 0, 0), \quad (1, 2, 3), \quad (3, 1, 2), \quad (7, 4, 7).$$

2007.1.2. Let

$$f(x) = \frac{e^x}{x}, \quad 1 \leq x \leq 2,$$

and let g be the inverse function of f . Evaluate

$$\int_e^{e^2/2} [g(x)]^2 dx.$$

2007.1.3. Let $A = \{1, 2, 3, \dots, n\}$. Suppose k distinct subsets of A , each containing exactly γ elements, are selected. Prove that if every element of A is to belong to at least p of the selected subsets, then

$$k \geq \frac{np}{\gamma}.$$

2007.1.4. Three unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ determine a parallelepiped of volume $\frac{1}{2}$. Let θ_{ij} be the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$, and let $\mathbf{A}$ be the 3×3 matrix whose (i, j) -entry is $\cos \theta_{ij}$. Compute $\det(\mathbf{A})$.

2007.1.5. Let $M_{2 \times 2}$ be the vector space of 2×2 matrices, and let $\mathbf{T} \in M_{2 \times 2}$ be invertible. Compute the determinant of the linear map

$$\Phi: M_{2 \times 2} \rightarrow M_{2 \times 2}, \quad \Phi(\mathbf{A}) = \mathbf{TAT}^{-1}.$$

2007.1.6. In triangle ABC , let P be a point on the side BC . Put

$$BP = x, \quad BC = l,$$

let the altitude of the triangle be h , and let

$$\angle APC = \theta(x).$$

Prove that

$$\angle A = \frac{1}{h} \int_0^l \sin^2 \theta(x) \, dx.$$

2007.1.7. Let $\mathbf{A}$ and $\mathbf{B}$ be real matrices. Suppose there exists a complex matrix $\mathbf{S}$ such that

$$\mathbf{A} = \mathbf{S}^{-1} \mathbf{B} \mathbf{S}.$$

Prove that there exists a real matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{R}^{-1} \mathbf{B} \mathbf{R}.$$

2007.1.8. Let $a_1 > 0$ and

$$a_{n+1} = \ln(1 + a_n).$$

Prove that

$$\lim_{n \rightarrow \infty} n a_n = 2.$$

2007.1.9. Let (a_n) be a sequence such that

$$\sum_{n=1}^{\infty} a_n = 1.$$

Prove that

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k \left(1 - \frac{k^2}{n^2} \right) = 1.$$

2007.1.10. Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfy the following condition: for every triangle Δ of area 1, $F(\Delta)$ is also a triangle of area 1. Prove that for every straight line l , $F(l)$ is also a straight line.

2008.2.1. Find the volume of the tetrahedron whose vertices are

$$(0, 1, 0), \quad (1, 2, 1), \quad (1, 3, 3), \quad (3, 1, 2).$$

2008.2.2. Evaluate

$$\int_0^{\infty} \frac{e^{-x} - e^{-2x}}{x} \, dx.$$

2008.2.3. Evaluate

$$\int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} xyz \cos(x^2 + y^2 + z^2) \, dx \, dy \, dz.$$

2008.2.4. Let $\alpha > 1$. Prove that the improper integral

$$\int_1^{\infty} \cos(x^\alpha) \, dx$$

converges.

2008.2.5. Find one matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

2008.2.6. Find a function $f(x, y, z)$ satisfying

$$\nabla f(x, y, z) = \left(ye^{xy+yz}, (x+z)e^{xy+yz}, ye^{xy+yz} \right).$$

2008.2.7. Find the maximum value of

$$f(x, y) = 3x^2 + 4xy + y^2$$

on the set

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

2008.2.8. Find all 3×1 column vectors $\mathbf{u}$ and $\mathbf{v}$ satisfying

$$\begin{pmatrix} 5 & 2 & 3 \\ 15 & 6 & 9 \\ 10 & 4 & 6 \end{pmatrix} = \mathbf{u}\mathbf{v}^\top.$$

2008.2.9. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix satisfying

$$\sum_{i=1}^n a_{ij} = 0$$

for every $j = 1, 2, \dots, n$. Prove that $\mathbf{A}$ is not invertible.

2008.2.10. Let (ϕ_n) be a sequence of functions defined on $[0, 1]$ satisfying

(a) $\phi_0(t) = 0$;

(b)

$$\phi_n(t) = 1 + 2 \sin \frac{t}{2} - \frac{1}{2} \int_0^t \phi_{n-1}(s) \, ds, \quad n = 1, 2, \dots$$

Suppose that $\phi_n \rightarrow \phi$ on $[0, 1]$ and that ϕ is differentiable. Find $\phi(t)$.

2008.2.11. Find all polynomial functions $u(x, y)$ satisfying the following conditions.

(a) $u_{xx} + u_{yy} = 0$.

(b) For every $x \in \mathbb{R}$,

$$u(x, 0) = u(x, \sqrt{3}x) = 0.$$

(c) For every $x > 0$ and $0 < y < \sqrt{3}x$,

$$u(x, y) > 0.$$

(d) There exists a positive integer n such that, for every $t \in \mathbb{R}$,

$$u(tx, ty) = t^n u(x, y).$$

2008.2.12. For every $n = 0, 1, 2, 3, \dots$, let a_n be either 0 or 1. Let P and Q be polynomials with $\deg P < \deg Q$, and suppose that for $|x| < 1$,

$$\sum_{n=0}^{\infty} a_n x^n = \frac{P(x)}{Q(x)}.$$

Prove that there exists a positive integer p such that

$$a_{n+p} = a_n$$

for every $n \geq 0$.

2008.2.13. For the four data points

$$(0, 0), \quad (1, 1), \quad (3, 2), \quad (4, 5),$$

find the line $y = ax + b$ that minimizes the sum of squared errors

$$E^2 = \sum_{i=1}^4 (y_i - ax_i - b)^2.$$

2008.1.1. Find the volume of the tetrahedron whose vertices are

$$(0, 1, 0), \quad (1, 2, 1), \quad (1, 3, 3), \quad (3, 2, 1).$$

2008.1.2. Let $0 < a, b, c, d < 1$. Prove the inequality

$$\begin{aligned} \sqrt{(1-ab)(1-cd)} &\geq \sqrt{ac(1-b)(1-d)} + \sqrt{bd(1-a)(1-c)} \\ &\quad + \sqrt{(1-a)(1-b)(1-c)(1-d)}. \end{aligned}$$

2008.1.3. Evaluate

$$\int_0^{\infty} \frac{e^{-x} - e^{-2x}}{x} dx.$$

2008.1.4. Let $\alpha > 1$. Prove that the improper integral

$$\int_1^{\infty} \cos(x^\alpha) dx$$

converges.

2008.1.5. Evaluate

$$\sin \frac{\pi}{10} \sin \frac{2\pi}{10} \cdots \sin \frac{9\pi}{10}.$$

2008.1.6. Let $\mathbf{A}$ be a 3×3 matrix all of whose entries are positive, and suppose that $\mathbf{A}^{-1}$ exists. If exactly six entries of $\mathbf{A}^{-1}$ are positive, prove that exactly three entries of $\mathbf{A}^{-1}$ are negative.

2008.1.7. Let (ϕ_n) be a sequence of functions defined on $[0, 1]$ satisfying

- (a) $\phi_0(t) = 0$;
- (b)

$$\phi_n(t) = 1 + 2 \sin \frac{t}{2} - \frac{1}{2} \int_0^t \phi_{n-1}(s) ds, \quad n = 1, 2, \dots$$

Show that (ϕ_n) converges on $[0, 1]$, and determine

$$\phi = \lim_{n \rightarrow \infty} \phi_n.$$

2008.1.8. For every $n = 0, 1, 2, 3, \dots$, let a_n be either 0 or 1. Let P and Q be polynomials with $\deg P < \deg Q$, and suppose that for $|x| < 1$,

$$\sum_{n=0}^{\infty} a_n x^n = \frac{P(x)}{Q(x)}.$$

Prove that there exists a positive integer p such that

$$a_{n+p} = a_n$$

for every $n \geq 0$.

2008.1.9. Let $p_i, q_i > 0$ and suppose that

$$\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1.$$

Prove that

$$\frac{1}{2} \sum_{i=1}^n (p_i - q_i)^2 \leq \sum_{i=1}^n p_i \ln \left(\frac{p_i}{q_i} \right).$$

2008.1.10. Let (x_n) be a sequence satisfying

$$x_m > 0, \quad x_{m+n} \leq x_m + x_n, \quad m, n = 1, 2, 3, \dots$$

Prove that the sequence (x_n/n) converges.

2009.2.1. Find the length of the curve

$$y = \frac{1}{2}x^2, \quad -1 \leq x \leq 1.$$

2009.2.2. Find the area of the region in the plane enclosed by

$$x^{2/3} + y^{2/3} = 1.$$

2009.2.3. Let

$$\mathbf{A} = \begin{pmatrix} -3 & 1 & 4 \\ -4 & 2 & 4 \\ -1 & 1 & 2 \end{pmatrix}.$$

Find the sum of all entries of $\mathbf{A}^{2009}$.

2009.2.4. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be nonzero column vectors, and suppose that

$$\mathbf{A} = \mathbf{xy}^{\top}$$

is symmetric. Prove that there exists a vector $\mathbf{u} \in \mathbb{R}^n$ such that

$$\mathbf{A} = \mathbf{uu}^{\top} \quad \text{or} \quad \mathbf{A} = -\mathbf{uu}^{\top}.$$

Here $\mathbf{y}^{\top}$ denotes the transpose of $\mathbf{y}$.

2009.2.5. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the following conditions.

(i) $f(x) > 0$ for every $x \in \mathbb{R}$.

(ii)

$$f'(x) - 6f(x)f'(x) - f'''(x) = 0$$

for every $x \in \mathbb{R}$.

(iii)

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} f'(x) = \lim_{x \rightarrow \infty} f''(x) = 0.$$

2009.2.6. Evaluate

$$\int_{-\pi}^{\pi} \frac{x^2}{1 + \sin x + \sqrt{1 + \sin^2 x}} dx.$$

2009.2.7. Prove that, for every positive integer n ,

$$\left(\frac{2}{3}\right)^{2n} + \left(\frac{2}{5}\right)^{2n} + \left(\frac{2}{7}\right)^{2n} + \cdots < \frac{1}{n}.$$

2009.2.8. Let $f: [-\pi, \pi] \rightarrow \mathbb{R}$ be integrable and suppose that

$$\int_{-\pi}^{\pi} f(x)^2 dx < \infty.$$

For each nonnegative integer k , define

$$a_k = \int_{-\pi}^{\pi} f(x) \cos(kx) dx, \quad b_k = \int_{-\pi}^{\pi} f(x) \sin(kx) dx.$$

Prove that

$$\lim_{k \rightarrow \infty} a_k = \lim_{k \rightarrow \infty} b_k = 0.$$

2009.1.1. Evaluate

$$\int_0^1 \int_y^1 \frac{\cos(\ln(x^2 + 1))}{x^2 + 1} dx dy.$$

2009.1.2. Evaluate

$$\int_{-\pi}^{\pi} \frac{x^2}{1 + \sin x + \sqrt{1 + \sin^2 x}} dx.$$

2009.1.3. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the following conditions.

(i) $f(x) > 0$ for every $x \in \mathbb{R}$.

(ii)

$$f'(x) - 6f(x)f'(x) - f'''(x) = 0$$

for every $x \in \mathbb{R}$.

(iii)

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} f'(x) = \lim_{x \rightarrow \infty} f''(x) = 0.$$

2009.1.4. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ be nonzero column vectors, and suppose that

$$\mathbf{A} = \mathbf{x}\mathbf{y}^\top$$

is symmetric. Prove that there exists a vector $\mathbf{u} \in \mathbb{R}^n$ such that

$$\mathbf{A} = \mathbf{u}\mathbf{u}^\top \quad \text{or} \quad \mathbf{A} = -\mathbf{u}\mathbf{u}^\top.$$

Here $\mathbf{y}^\top$ denotes the transpose of $\mathbf{y}$.

2009.1.5. Let $p \geq 5$ be prime. Express

$$\sum_{k=1}^{p^2} \frac{1}{k} - \frac{1}{p} \sum_{k=1}^p \frac{1}{k}$$

as a reduced fraction. Prove that its numerator is divisible by p^4 .

2009.1.6. Let f be twice differentiable on $[0, 2]$ and suppose that

$$f''(x) \geq 0 \quad (0 \leq x \leq 2).$$

Prove that

$$\int_0^2 f(x) \, dx \geq 2f(1).$$

2009.1.7. For a real number a and subsets $A, B \subseteq \mathbb{R}$, define

$$aA = \{ax \mid x \in A\}, \quad A + B = \{a + b \mid a \in A, b \in B\}.$$

For each positive integer n , define

$$G_0 = [0, 1) = \{x \mid 0 \leq x < 1\}, \quad G_n = \frac{1}{5}(\{0, 2, 4\} + G_{n-1}).$$

Prove that the set

$$G = \bigcap_{n=1}^{\infty} G_n$$

is uncountable.

2009.1.8. Let

$$g(x) = \frac{1}{3}x^2 + \frac{2}{3}x.$$

Suppose that a continuous function $f: [0, 1] \rightarrow [0, 1]$ satisfies

$$|f(x) - f(y)| \leq g(|x - y|)$$

for all $x, y \in [0, 1]$. Prove that the set

$$A = \{c \in [0, 1] \mid f(c) = g(c)\}$$

either consists of a single point or is a closed interval.

2010.2.1. Evaluate

$$\int_0^\infty \frac{1}{x^3 + 1} dx.$$

2010.2.2. Find the surface area obtained by rotating the curve

$$y = \frac{x^2}{2}, \quad 0 \leq x \leq 1,$$

about the x -axis.

2010.2.3. Find the general solution on $x > 1$ of the differential equation

$$(x-1)y'' - xy' + y = 0.$$

2010.2.4. Let $n \geq 3$. Solve the system

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ 2 & 3 & 4 & \cdots & 1 \\ 3 & 4 & 5 & \cdots & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n & 1 & 2 & \cdots & n-1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} n \\ n-1 \\ n-2 \\ \vdots \\ 1 \end{pmatrix}.$$

2010.2.5. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices satisfying

$$\mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

Prove that

$$\mathbf{AB} = \mathbf{BA}.$$

2010.2.6. Let $y = f(x)$ be a continuous function having a local extremum at $x = 1$. Suppose that, in a neighborhood of $x = 1$,

$$y^3 + 3xy^2 + x^3y = 1.$$

Find the second-degree Taylor polynomial of f at $x = 1$, and determine whether $f(1)$ is a local maximum or a local minimum.

2010.2.7.

(a) Find a cubic polynomial $f(x)$ with integer coefficients such that

$$f\left(\cos \frac{\pi}{7}\right) = 0.$$

(b) Prove that

$$4 \sin \frac{2\pi}{7} - \tan \frac{\pi}{7} = \sqrt{7}.$$

2010.2.8. For $a \in \mathbb{R}$, evaluate

$$\int_0^\infty e^{-x^2} \cos(ax) dx.$$

2010.1.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and satisfy

$$\sum_{k=0}^{2009} f(x+k) = x^{2009}$$

for every $x \in \mathbb{R}$. Evaluate

$$\int_0^{2010} f(x) \, dx.$$

2010.1.2. Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Find the minimal polynomial of $\mathbf{A}$, and prove that there is no matrix $\mathbf{B}$ satisfying

$$\mathbf{B}^2 = \mathbf{A}.$$

2010.1.3. Let

$$f(x) = x^m + a_{m-1}x^{m-1} + a_{m-2}x^{m-2} + \cdots + a_1x + a_0$$

divide $x^n - 1$, and let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{m-1} \end{pmatrix}.$$

Prove that

$$\det(x^{n-1}\mathbf{I}_m + x^{n-2}\mathbf{A} + x^{n-3}\mathbf{A}^2 + \cdots + x^2\mathbf{A}^{n-3} + x\mathbf{A}^{n-2} + \mathbf{A}^{n-1}) = \frac{(x^n - 1)^m}{f(x)}.$$

2010.1.4. Let p be a prime such that $p-1$ is not divisible by 7. Prove that there are infinitely many positive integers n for which

$$p \mid n^7 - 2010.$$

2010.1.5. Find all real numbers a for which the improper integral

$$\int_0^\infty \frac{\sin^2 x}{x^a} \, dx$$

converges.

2010.1.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ have a continuous derivative. For $x \neq y$, define

$$g(x, y) = \frac{f(x) - f(y)}{x - y}.$$

Prove that $g(x, x)$ can be defined so that g is continuous on $\mathbb{R}^2$.

2010.1.7. Evaluate

$$\int_{1/2}^2 \frac{\arctan x}{x^2 - x + 1} dx.$$

2010.1.8. Find positive real numbers a and r satisfying

$$\lim_{n \rightarrow \infty} n^r \cdot \frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} = a.$$

2011.2.1. Let

$$f(x) = \frac{1}{x - x^{3/5}}.$$

Evaluate

$$\int_{2^5}^{3^5} f(x) dx.$$

2011.2.2. For an arbitrary $n \times n$ real matrix $\mathbf{A}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{A}^2 \\ \mathbf{A}^3 & \mathbf{A}^4 \end{pmatrix} = 0.$$

2011.2.3. Let

$$f(x) = e^{\cos(x^2)}.$$

Find

$$f^{(8)}(0).$$

2011.2.4. Let (y_n) be defined by

$$y_1 = 1, \quad y_2 = 3,$$

and, for every positive integer n ,

$$y_{n+2} = \frac{2}{n} \left((n^2 + 3n + 1)y_{n+1} - 2(n+1)^2 y_n \right).$$

Prove that every term of the sequence is a positive integer.

2011.2.5. Let f be a cubic polynomial. Prove that, for every positive integer n ,

$$\begin{aligned} \int_0^{2n} f(x) dx &= \frac{1}{3} (f(0) + 4f(1) + 2f(2) + 4f(3) + 2f(4) + \cdots \\ &\quad + 2f(2n-2) + 4f(2n-1) + f(2n)). \end{aligned}$$

2011.2.6. Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} i^2 - 1, & i = j, \\ ij, & i \neq j. \end{cases}$$

Find all eigenvalues of $\mathbf{A}$ and compute $\det(\mathbf{A})$.

2011.2.7. For a continuous function $f: [0, 1] \rightarrow [0, \infty)$, define

$$I(f) = \int_0^1 (x^2 f(x) - f(x)^3) \, dx.$$

Find the maximum possible value of $I(f)$.

2011.2.8. Let D be the closed unit disk centered at the origin in the coordinate plane.

(a) Evaluate

$$\iint_D \frac{1}{\sqrt{1-x^2-y^2}} \, dx \, dy.$$

(b) Suppose that n rectangular pieces of paper are given, each of width 2011 and with respective heights $a_1, \dots, a_n$. Prove that if

$$\sum_{i=1}^n a_i < 2,$$

then these pieces cannot cover D . Overlap is allowed, but folding and tearing are not.

2011.1.1. Define

$$f(x) = \int_{\sin x}^{1+x} e^{t^2+2xt} \, dt.$$

Find $f'(0)$.

2011.1.2. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix all of whose entries are positive real numbers, and suppose that

$$\sum_{i=1}^n a_{ij} = 1$$

for every $j = 1, 2, \dots, n$. Prove that every root of

$$\det(\mathbf{A} - x\mathbf{I}_n) = 0$$

has absolute value at most 1.

2011.1.3. Evaluate

$$\int_0^\infty x^3 \pi^{-\sqrt[3]{x^8}} \, dx.$$

2011.1.4. For each positive integer n , let

$$A(n) = \left| \{(x, y) \in \mathbb{Z}^2 \mid x^2 + 3y^2 = n\} \right|.$$

Prove that

$$\left| \{(x, y) \in \mathbb{Z}^2 \mid x^2 + 3(4y^2 - 3y) = n\} \right| = \frac{1}{2} A(16n + 27).$$

2011.1.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 |f'(x)|^2 \, dx \geq \frac{1}{4} \int_0^1 \frac{|f(x)|^2}{x^2} \, dx.$$

2011.1.6. Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} i^2 - 1, & i = j, \\ ij, & i \neq j. \end{cases}$$

Find all eigenvalues of $\mathbf{A}$ and compute $\det(\mathbf{A})$.

2011.1.7. Let D be the closed unit disk centered at the origin in the coordinate plane.

(a) Evaluate

$$\iint_D \frac{1}{\sqrt{1-x^2-y^2}} \, dx \, dy.$$

(b) Suppose that n rectangular pieces of paper are given, each of width 2011 and with respective heights $a_1, \dots, a_n$. Prove that if

$$\sum_{i=1}^n a_i < 2,$$

then these pieces cannot cover D . Overlap is allowed, but folding and tearing are not.

2011.1.8. Let a line ℓ and a plane P in three-dimensional Euclidean space meet at the origin. Let f be a rotation about the axis ℓ , and let g be reflection across the plane P . Prove that there exist a line ℓ' and a plane P' perpendicular to each other, together with a rotation f' about the axis ℓ' and a reflection g' across P' , such that

$$f \circ g = f' \circ g'.$$

2012.2.1. Let

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -4 & 1 & 4 \end{pmatrix}.$$

Find the sum of all entries of $\mathbf{A}^{2012}$.

2012.2.2. Find all continuous functions $f: [0, 1] \rightarrow [0, \infty)$ satisfying

$$\int_0^1 x f(x) \, dx = 3 \int_0^1 x^2 f(x) \, dx = 9 \int_0^1 x^3 f(x) \, dx.$$

2012.2.3. The region in the coordinate plane bounded by the line $y = x$ and the curve $y = x^2$ is rotated about the line $y = x$. Find the volume of the resulting solid.

2012.2.4. Let

$$P = (0, 1), \quad Q = (2\sqrt{3}, 3).$$

For a point X on the x -axis, find the minimum value of

$$|X - P| + \sqrt{3}|X - Q|.$$

2012.2.5. Find all possible values of $\text{tr}(\mathbf{A})$ for $n \times n$ matrices $\mathbf{A}$ satisfying the following conditions, where $\mathbf{I}_n$ is the identity matrix.

(i) $\mathbf{A}^2 = \mathbf{I}_n$.

(ii) $\text{rank}(\mathbf{A} + \mathbf{I}_n) = 1$.

2012.2.6. For real numbers x, y, z , define

$$m_k = \frac{1}{3}(x^k + y^k + z^k).$$

If $m_1 = 0$, prove that

$$m_3^2 \leq m_2(m_4 - m_2^2).$$

2012.2.7. Define

$$e_n = \left(1 + \frac{1}{n}\right)^n.$$

Evaluate

$$\lim_{n \rightarrow \infty} \left(\frac{2n(e - e_n)}{e} \right)^n,$$

where e is the base of the natural logarithm.

2012.2.8. Let $\mathbf{A}$ be a real symmetric matrix satisfying the following conditions.

- (i) Every diagonal entry of $\mathbf{A}$ is 0.
- (ii) Some entry of $\mathbf{A}$ is greater than 1.

Prove that $\mathbf{A}$ has an eigenvalue greater than 1.

2012.1.1. Evaluate

$$J = \int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} \, dx.$$

2012.1.2. Prove that there do not exist square matrices $\mathbf{A}$ and $\mathbf{B}$ satisfying

$$(\mathbf{AB})^{2012} - (\mathbf{BA})^{2012} = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix.

2012.1.3. Evaluate the right-hand limit

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{\tan x - x}}{\sin \sqrt{x} - \sqrt{x}}.$$

2012.1.4. For a fixed integer $n \geq 2$, find all possible values of $\text{tr}(\mathbf{A})$ for $n \times n$ matrices $\mathbf{A}$ satisfying the following conditions, where $\mathbf{I}_n$ is the identity matrix.

- (i) $\text{rank}(\mathbf{A} + \mathbf{I}_n) = 1$.
- (ii) $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^3)$.

2012.1.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that

$$2 \int_0^{1/2} f(x) \, dx = \int_{1/2}^1 f(x) \, dx.$$

Prove that

$$3 \int_0^1 (f'(x))^2 \, dx \geq \left(2 \int_0^1 f(x) \, dx \right)^2.$$

2012.1.6. Let $\mathbf{A}$ be a square matrix with integer entries satisfying the following conditions, where $\mathbf{I}$ is the identity matrix.

- (i) For an odd prime p , every entry of $\mathbf{A} - \mathbf{I}$ is divisible by p .
- (ii) There exists a positive integer k such that $\mathbf{A}^k = \mathbf{I}$.

Prove that $\mathbf{A} = \mathbf{I}$.

2012.1.7. For a real sequence $X = (x_i)_{i=1}^n$, define

$$m_k(X) = \frac{1}{n} \sum_{i=1}^n x_i^k.$$

If $m_1(X) = 0$, prove that

$$(m_3(X))^2 \leq m_2(X) \left(m_4(X) - (m_2(X))^2 \right).$$

2012.1.8. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable, with $f(0) = 0$, and suppose that

$$(f'(x))^2 \leq f(x)f''(x) \quad (x \in [0, 1]).$$

Prove that $f(1) = 0$.

2013.2.1. Evaluate the right-hand limit

$$\lim_{a \rightarrow 0^+} \sqrt{a} \iiint_{\mathbb{R}^3} \frac{1}{\left(\sqrt{x^2 + y^2 + z^2} + a \right)^{7/2}} dx dy dz.$$

2013.2.2. Let $n \geq 2$ be an integer and let $a, b \in \mathbb{R}$. Let $\mathbf{M}_n(a, b)$ be the $n \times n$ matrix whose diagonal entries are all a and whose off-diagonal entries are all b . Find

$$\det(\mathbf{M}_n(a, b)).$$

2013.2.3. In $\mathbb{R}^3$, let

$$\begin{aligned} A &= (2, 0, -1), & B &= (-1, 3, -1), & C &= (-3, 1, 1), \\ P &= (-2, 1, 1), & Q &= (1, 1, -2). \end{aligned}$$

- (a) Prove that the line segment PQ meets the interior of triangle ABC .
- (b) Find the volume of the convex hexahedron with vertices A, B, C, P, Q whose faces are all triangles.

2013.2.4. Prove that

$$\int_{\mathbb{R}} \frac{x^2}{x^4 + 1} dx = \int_{\mathbb{R}} \frac{1}{x^4 + 1} dx,$$

and find their common value.

2013.2.5. For arbitrary $n \times n$ square matrices $\mathbf{A}, \mathbf{B}$ and the $n \times n$ identity matrix $\mathbf{I}_n$, prove that

$$\det(\mathbf{I}_n + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

2013.2.6. Choose four distinct points at random on the unit circle and form the quadrilateral having these points as its vertices. Find the average area of such quadrilaterals.

2013.2.7. Let $\mathbf{A}$ be an $n \times n$ real matrix whose row vectors all have norm 1. Prove that if

$$\operatorname{tr}(\mathbf{A}) > \sqrt{n(n-1)},$$

then

$$\det(\mathbf{A}) \neq 0.$$

2013.2.8. Prove that there exists a positive constant C , independent of x , such that

$$\sum_{n=1}^{\infty} \frac{|x|}{n^2(|n-x|+1)^2} < C$$

for every $x \in \mathbb{R}$.

2013.1.1. Let $\mathbf{A}$ be a 2013×2013 matrix such that each row consists of distinct positive integers not exceeding 2013. Prove that $\det(\mathbf{A})$ is divisible by 2013.

2013.1.2. Find the volume of the region

$$\{(x, y, z) \in \mathbb{R}^3 \mid (x^2 + y^2 + 4z^2 + 3)^2 \leq 16(x^2 + y^2)\}.$$

2013.1.3. For each positive integer n , let x_n be the unique nonnegative real solution of

$$x^n + x^{n-1} + x - 1 = 0.$$

Prove that the sequence (x_n) is increasing and converges to 1.

2013.1.4. For arbitrary $n \times n$ real matrices $\mathbf{A}, \mathbf{B}$ and the $n \times n$ identity matrix $\mathbf{I}_n$, prove that

$$\operatorname{rank}(\mathbf{I}_n + \mathbf{AB}) = \operatorname{rank}(\mathbf{I}_n + \mathbf{BA}).$$

2013.1.5. Define

$$S = \{n \in \mathbb{Z} \mid 2x^2 + xy + 4y^2 = n \text{ for some } x, y \in \mathbb{Z}\}.$$

Prove the following.

- (a) If $m \in S$, then $2m \in S$.
- (b) If $m, n \in S$, then $mn \in S$.

2013.1.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and suppose that, for every $x \in [0, 1]$,

$$f(x) \geq 0, \quad f''(x) \leq 0.$$

Prove that

$$\int_0^1 xf(x) \, dx \leq \frac{2}{3} \int_0^1 f(x) \, dx.$$

2013.1.7. Let $\mathbf{A}$ be an $n \times n$ real matrix whose row vectors all have norm 1. Prove that if

$$\operatorname{tr}(\mathbf{A}) > \sqrt{n(n-1)},$$

then

$$\det(\mathbf{A}) \neq 0.$$

2013.1.8. Let $f: [1, \infty) \rightarrow (1, \infty)$ be monotonically increasing and suppose that, for every $x \geq 1$,

$$f(x)^2 \leq f(4x) \leq 2013^{\sqrt{x}}.$$

Evaluate

$$\lim_{x \rightarrow \infty} \frac{\ln \ln f(x)}{\ln x}.$$

2014.2.1. For a positive integer n , let

$$f(x) = \frac{e^x}{x-1},$$

and define the polynomial $P_n(x)$ by

$$f^{(n)}(x) = \frac{P_n(x)e^x}{(x-1)^{n+1}}.$$

Find $P_n(1)$ and $P'_n(1)$.

2014.2.2. Let (a_n) and (b_n) be real sequences satisfying

$$\lim_{n \rightarrow \infty} (a_n + b_n) = 0, \quad \lim_{n \rightarrow \infty} (e^{a_n} + e^{b_n}) = 2.$$

Prove that (a_n) converges.

2014.2.3. Let m be a positive integer. Prove that every $n \times n$ real matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^m = \mathbf{0}$$

also satisfies

$$\operatorname{rank}(\mathbf{A}) \leq \frac{m-1}{m}n.$$

2014.2.4. Evaluate

$$\iiint_{\mathbb{R}^3} e^{-3x^2-3y^2-2z^2-2xz+2yz} \, dx \, dy \, dz.$$

2014.2.5. Let $n \geq 2$ be an integer, and let $M_n(\mathbb{R})$ denote the real vector space of all $n \times n$ real matrices. Let $V \subseteq M_n(\mathbb{R})$ satisfy the following conditions, where $\mathbf{E}_{ij}$ denotes the matrix whose (i, j) -entry is 1 and whose other entries are 0.

- (i) V is a subspace of $M_n(\mathbb{R})$.
- (ii) If $\mathbf{A}, \mathbf{B} \in V$, then $\mathbf{AB} \in V$.
- (iii)

$$\mathbf{E}_{12} + \mathbf{E}_{23} + \cdots + \mathbf{E}_{n-1,n} \in V,$$

and

$$\mathbf{E}_{21} + \mathbf{E}_{32} + \cdots + \mathbf{E}_{n,n-1} \in V.$$

Prove that

$$V = M_n(\mathbb{R}).$$

2014.2.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 |f(x)f'(x)| \, dx \leq \frac{1}{2} \int_0^1 |f'(x)|^2 \, dx.$$

2014.2.7. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices all of whose eigenvalues are positive. Prove that, for every $n \times n$ real matrix $\mathbf{C}$, the block matrix

$$\begin{pmatrix} \mathbf{A} & \mathbf{C} \\ \mathbf{C}^\top & -\mathbf{B} \end{pmatrix}$$

is invertible. Here $\mathbf{C}^\top$ denotes the transpose of $\mathbf{C}$.

2014.2.8. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous, with $a < b$, and suppose that

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for every $t \in [0, 1]$ and $x, y \in [a, b]$. Prove that

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) \, dx \leq \frac{f(a) + f(b)}{2}.$$

2014.1.1. Evaluate

$$\lim_{n \rightarrow \infty} \left(\frac{1}{n!} \right)^{1/(n \ln n)}.$$

2014.1.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices, and for $t \in \mathbb{R}$ define

$$\mathbf{C}_t = \begin{pmatrix} \mathbf{A} & t^2 \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix}.$$

Prove that

$$\det(\mathbf{C}_t) = \det(\mathbf{A} + t\mathbf{B}) \det(\mathbf{A} - t\mathbf{B}).$$

2014.1.3. Find all continuous functions $f: [0, \infty) \rightarrow (0, \infty)$ satisfying

$$f(x)f(y) = \max\{f(t) \mid |x-y| \leq t \leq x+y\}$$

for all $x, y \geq 0$.

2014.1.4. Let $M_n(\mathbb{C})$ be the complex vector space of all $n \times n$ complex matrices. Suppose $\mathbf{A} \in M_n(\mathbb{C})$ satisfies

$$\mathbf{A}^6 - \mathbf{A}^3 + \mathbf{I}_n = \mathbf{0}.$$

Define the linear map

$$T: M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C}), \quad T(\mathbf{B}) = \mathbf{A}\mathbf{B}.$$

Prove that T is diagonalizable.

2014.1.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable and satisfy $f(0) = 0$. Prove that

$$\int_0^1 |f(x)f'(x)| \, dx \leq \frac{1}{2} \int_0^1 |f'(x)|^2 \, dx.$$

2014.1.6. Let n be a positive integer, and let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices satisfying

$$x\mathbf{A} + y\mathbf{B} = \mathbf{I}_n, \quad \mathbf{AB} = \mathbf{0},$$

where $x, y \in \mathbb{R} \setminus \{0\}$. Prove that

$$\det(\mathbf{A} + \mathbf{B}) = \frac{1}{x^{\text{rank}(\mathbf{A})}y^{\text{rank}(\mathbf{B})}}.$$

2014.1.7. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous, with $a < b$, and suppose that

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$

for every $t \in [0, 1]$ and $x, y \in [a, b]$. Prove that

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f(x) \, dx \leq \frac{f(a) + f(b)}{2}.$$

2014.1.8. Let α be a positive irrational number and define

$$q_n = \frac{\lfloor n\alpha \rfloor}{n} \quad (n \geq 1).$$

Prove that the sequence (q_n) is not monotonically increasing. Here $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2015.2.1. Evaluate

$$\int_0^\infty (x^2 + 1 - x\sqrt{x^2 + 2}) \, dx.$$

2015.2.2. For a positive integer n , let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \max\{i, j\}.$$

Find $\det(\mathbf{A})$.

2015.2.3. Determine whether the series

$$\sum_{n=1}^{\infty} \left((2n+1)e^{-1/n} - (2n-1) \right)$$

converges.

2015.2.4. Let $\mathbf{A}$ be a 5×5 matrix with rational entries having the eigenvector

$$(1, \sqrt{2}, \sqrt{3}, \sqrt{4}, \sqrt{5})^T.$$

Prove that $\mathbf{A}^T$ has an eigenvector all of whose entries are rational. Here $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$.

2015.2.5. Let $f: (-1, \infty) \rightarrow \mathbb{R}$ be infinitely differentiable and suppose that, for every positive integer n ,

$$f\left(\frac{1}{n}\right) = \frac{n}{n+1}.$$

Find $f'''(0)$.

2015.2.6. Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Prove that there exists a constant $c \in [a, b]$ such that

$$\int_a^b f(x)e^x \, dx = e^a \int_a^c f(x) \, dx + e^b \int_c^b f(x) \, dx.$$

2015.2.7. Let n be a positive integer, and let $\mathbf{A} = (a_{ij})$ be an $n \times n$ complex matrix satisfying the following conditions.

- (i) $a_{ii} = 1$ for $1 \leq i \leq n$.
- (ii) $a_{ij} = 0$ for $1 \leq j < i \leq n$.
- (iii) $\overline{\mathbf{A}}\mathbf{A} = \mathbf{I}_n$.

Prove that there exists an $n \times n$ complex matrix $\mathbf{B} = (b_{ij})$ satisfying the following conditions.

- (i) $b_{ii} = 1$ for $1 \leq i \leq n$.
- (ii) $b_{ij} = 0$ for $1 \leq j < i \leq n$.
- (iii) $\overline{\mathbf{B}}\mathbf{A} = \mathbf{B}$.

Here $\overline{\mathbf{A}} = (\overline{a_{ij}})$.

2015.2.8. Let n be a positive integer, and let $\mathbf{X} = (x_{ij})$ be an $n \times n$ real matrix whose entries vary over a domain on which $\mathbf{X}$ is invertible. Write

$$\mathbf{Y} = \mathbf{X}^{-1} = (y_{pq}),$$

and regard each entry as a function

$$y_{pq} = y_{pq}(x_{11}, x_{12}, \dots, x_{nn}).$$

Prove that

$$\frac{\partial y_{pq}}{\partial x_{ij}} = -y_{pi}y_{jq}.$$

2015.1.1. Evaluate

$$\lim_{n \rightarrow \infty} \left(1 + \frac{\ln n}{n-1}\right)^n.$$

2015.1.2. Let d be a positive integer and let

$$\phi(x, y) = \sum_{i=0}^{2d} a_i x^i y^{2d-i}$$

be an arbitrary polynomial with real coefficients. Prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^d \left(y \frac{\partial \phi}{\partial x}\right) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)^d \left(x \frac{\partial \phi}{\partial y}\right).$$

2015.1.3. Let $n \geq 3$ be an integer, and let $\mathbf{A} = (a_{ij})$ be the $n^2 \times n^2$ matrix defined by

$$a_{ij} = \begin{cases} 1, & n \mid i + j, \\ 0, & n \nmid i + j. \end{cases}$$

Find all eigenvalues of $\mathbf{A}$ and the dimension of the eigenspace corresponding to each eigenvalue.

2015.1.4. Let $V = M_{100}(\mathbb{R})$, the real vector space of all 100×100 real matrices. For $\mathbf{A} \in V$, let

$$d_{\mathbf{A}} = \dim\{\mathbf{B} \in V \mid \mathbf{AB} = \mathbf{BA}\}.$$

Suppose that $\mathbf{A} \in V$ satisfies

$$\mathbf{A}^4 - 5\mathbf{A}^2 + 4\mathbf{I} = \mathbf{0}.$$

Find the minimum possible value of $d_{\mathbf{A}}$.

2015.1.5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and satisfy

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x) = 0,$$

$$\int_{-\infty}^{\infty} |f(x)|^2 dx < \infty, \quad \int_{-\infty}^{\infty} |f'(x)|^2 dx < \infty.$$

Prove that

$$\max_{x \in \mathbb{R}} |f(x)| \leq \left(\int_{-\infty}^{\infty} |f(x)|^2 dx \right)^{1/4} \left(\int_{-\infty}^{\infty} |f'(x)|^2 dx \right)^{1/4}.$$

2015.1.6. Let $x(t), y(t), z(t)$ be relatively prime polynomials with real coefficients, each of degree at least 1. Suppose that, for some positive integer d ,

$$x(t)^d + y(t)^d = z(t)^d.$$

Prove that $x(t)^{d-1}$ divides

$$y(t)z'(t) - z(t)y'(t),$$

and that $d \leq 2$.

2015.1.7. Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 real symmetric matrices with eigenvalues

$$\lambda_1 \geq \lambda_2 \geq 0 \quad \text{and} \quad \mu_1 \geq \mu_2 \geq 0,$$

respectively. Prove that

$$\text{tr}(\mathbf{AB}) \leq \lambda_1 \mu_1 + \lambda_2 \mu_2.$$

2015.1.8. For a positive integer n , call an $n \times n$ real matrix $\mathbf{M}$ orthogonal if

$$\mathbf{M}^T \mathbf{M} = \mathbf{I}_n,$$

and call $\mathbf{M}$ similar to an orthogonal matrix if there exists an invertible real matrix $\mathbf{P}$ such that

$$\mathbf{PMP}^{-1}$$

is orthogonal.

- (a) Let $\mathbf{S}$ be a real symmetric matrix all of whose eigenvalues are positive. Prove that every matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^T \mathbf{S} \mathbf{A} = \mathbf{S}$$

is similar to an orthogonal matrix.

- (b) Let $\mathbf{A}$ and $\mathbf{B}$ be matrices such that both $\mathbf{A}$ and

$$\begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{pmatrix}$$

are similar to orthogonal matrices. Prove that $\mathbf{B}$ is also similar to an orthogonal matrix.

2016.2.1. Using $\ln$ for the natural logarithm, evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\sum_{k=2}^{2^n} k^{1/n^2} \right).$$

2016.2.2. Let $a, b \in \mathbb{R}$ satisfy

$$a^2 + b^2 \leq \frac{\pi^2}{4}.$$

Prove that

$$\cos \sqrt{a^2 + b^2} \leq \cos a \cos b.$$

2016.2.3. Let $\mathbf{A}$ be an $n \times n$ real matrix having n distinct positive eigenvalues. Find the number of real matrices $\mathbf{B}$ satisfying

$$\mathbf{B}^{2016} = \mathbf{A}.$$

2016.2.4. Let $y(t)$ satisfy

$$y'(t) = y(t) - \int_0^t (y'(s))^2 \cos s \, ds, \quad y(0) = 2.$$

Find $y(\pi/4)$.

2016.2.5. Let $f: [-\pi/4, \pi/4] \rightarrow [-1, 1]$ be continuous and differentiable on $(-\pi/4, \pi/4)$. Prove that there exists $x_0 \in (-\pi/4, \pi/4)$ such that

$$|f'(x_0)| \leq 1 + f(x_0)^2.$$

2016.2.6. Let m, n be positive integers, and let $\mathbf{A}$ be an $m \times n$ real matrix. Define

$$\mathbf{X} = \mathbf{I}_m + \mathbf{A}\mathbf{A}^\top, \quad \mathbf{Y} = \mathbf{I}_n + \mathbf{A}^\top\mathbf{A}.$$

(a) Prove that $\mathbf{X}$ and $\mathbf{Y}$ are invertible.

(b) Find

$$\text{tr}(\mathbf{X}^{-1}) - \text{tr}(\mathbf{Y}^{-1}).$$

Here $\mathbf{A}^\top$ denotes the transpose of $\mathbf{A}$, and $\mathbf{I}_\ell$ denotes the $\ell \times \ell$ identity matrix.

2016.2.7. Let $x_1, x_2, x_3 > 0$ and let $a \in [-\frac{1}{2}, 1]$. Define

$$m = \frac{x_1 + x_2 + x_3}{3},$$

$$y_1 = (1-a)m + ax_1, \quad y_2 = (1-a)m + ax_2, \quad y_3 = (1-a)m + ax_3.$$

Prove that

$$\frac{27y_1y_2y_3}{(y_1 + y_2 + y_3)^3} \geq \left(\frac{27x_1x_2x_3}{(x_1 + x_2 + x_3)^3} \right)^{\max\{a, -2a\}}.$$

2016.2.8. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real matrices satisfying

$$\mathbf{A} \neq \mathbf{0}, \quad \mathbf{AB} - \mathbf{BA} = \mathbf{A}.$$

Prove that

$$\mathbf{A}^n = \mathbf{0} \quad \text{and} \quad \mathbf{B}^n \neq \mathbf{0}.$$

2016.1.1. Using $\ln$ for the natural logarithm, evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\sum_{k=2}^{2^n} k^{1/n^2} \right).$$

2016.1.2. Let $(a_n)_{n \geq 1}$ be a decreasing real sequence satisfying

$$\lim_{n \rightarrow \infty} a_n = 0.$$

For

$$S_m = \sum_{n=1}^m a_n,$$

suppose that

$$S_{2^k} - 2^k a_{2^k} \leq 1$$

for every positive integer k . Prove that

$$\sum_{n=1}^{\infty} a_n \leq 1.$$

2016.1.3. Let m, n be positive integers, and let $\mathbf{A}$ be an $m \times n$ real matrix. Define

$$\mathbf{X} = \mathbf{I}_m + \mathbf{A}\mathbf{A}^\top, \quad \mathbf{Y} = \mathbf{I}_n + \mathbf{A}^\top \mathbf{A}.$$

- (a) Prove that $\mathbf{X}$ and $\mathbf{Y}$ are invertible.
- (b) Find

$$\text{tr}(\mathbf{X}^{-1}) - \text{tr}(\mathbf{Y}^{-1}).$$

Here $\mathbf{A}^\top$ denotes the transpose of $\mathbf{A}$, and $\mathbf{I}_\ell$ denotes the $\ell \times \ell$ identity matrix.

2016.1.4. Let $f: [-\pi/4, \pi/4] \rightarrow [-1, 1]$ be continuous and differentiable on $(-\pi/4, \pi/4)$. Prove that there exists $x_0 \in (-\pi/4, \pi/4)$ such that

$$|f'(x_0)| \leq 1 + f(x_0)^2.$$

2016.1.5. Let

$$f(x) = \cos \left(\frac{3\sqrt{3}\pi}{8} (x - x^3) \right).$$

Evaluate

$$\lim_{t \rightarrow \infty} \left(\int_0^1 f(x)^t dx \right)^{1/t} + \lim_{t \rightarrow -\infty} \left(\int_0^1 f(x)^t dx \right)^{1/t}.$$

2016.1.6. Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 real matrices satisfying

$$\det(\mathbf{A}) = \det(\mathbf{B}) = 1, \quad \text{tr}(\mathbf{A}) > 2, \quad \text{tr}(\mathbf{B}) > 2,$$

and

$$\text{tr}(\mathbf{A}\mathbf{B}\mathbf{A}^{-1}\mathbf{B}^{-1}) = 2.$$

Prove that $\mathbf{A}$ and $\mathbf{B}$ have a common eigenvector.

2016.1.7. Let $M > 0$, and let $f: [0, \infty) \rightarrow [0, M]$ be continuous and satisfy

$$\int_0^\infty (1+x)f(x) \, dx < \infty.$$

Prove that

$$\left(\int_0^\infty f(x) \, dx \right)^2 \leq 4M \int_0^\infty xf(x) \, dx.$$

2016.1.8. Let $\mathbf{A}$ be an $n \times n$ complex matrix. Prove that the following conditions are equivalent.

(i) There exists an $n \times n$ complex matrix $\mathbf{B}$ such that

$$\mathbf{AB} - \mathbf{BA} = \mathbf{A}.$$

(ii) There exists a positive integer k such that

$$\mathbf{A}^k = \mathbf{0}.$$